ENGINEERING PORTFOLIO

Selected work From my SERC Internship

Tami Omole

CAD · analysis · prototyping

Two projects

01 Optical Reef

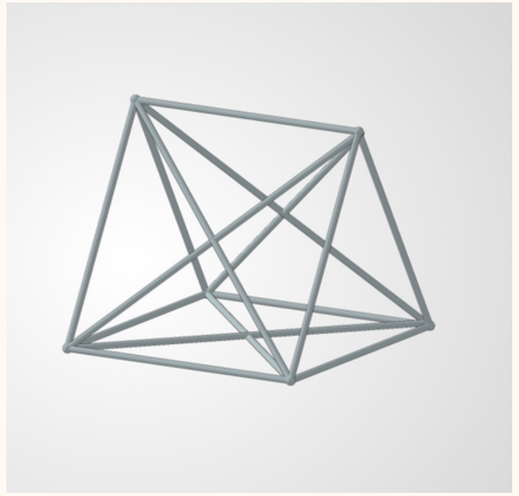
Parameterized CAD, patterned assemblies, frequency analysis.

02 Floatbots

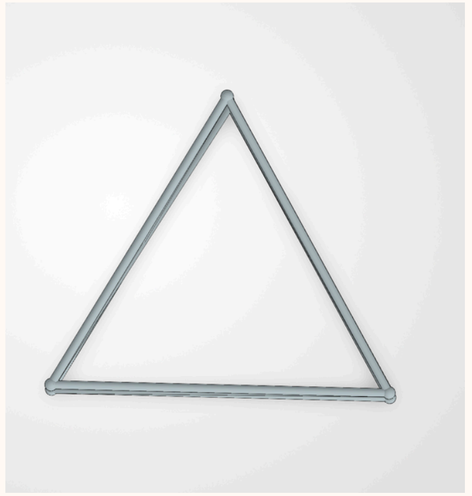
An aluminum frame, adjustable camera mount, and combined electronics enclosure.

One node Optical Reef Truss

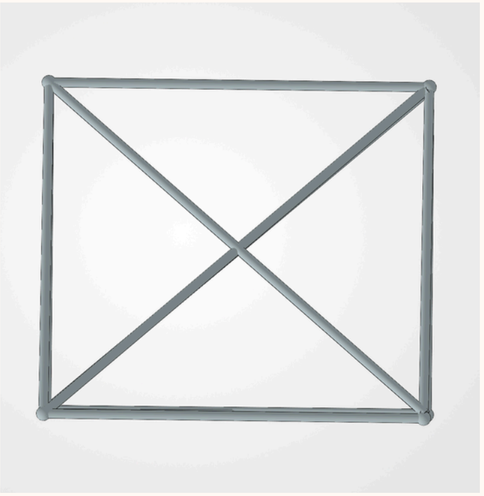
I modeled one node in Siemens NX using tubes joined at spheres. The main dimensions were parameterized so the same geometry could be reused.



ISOMETRIC



FRONT

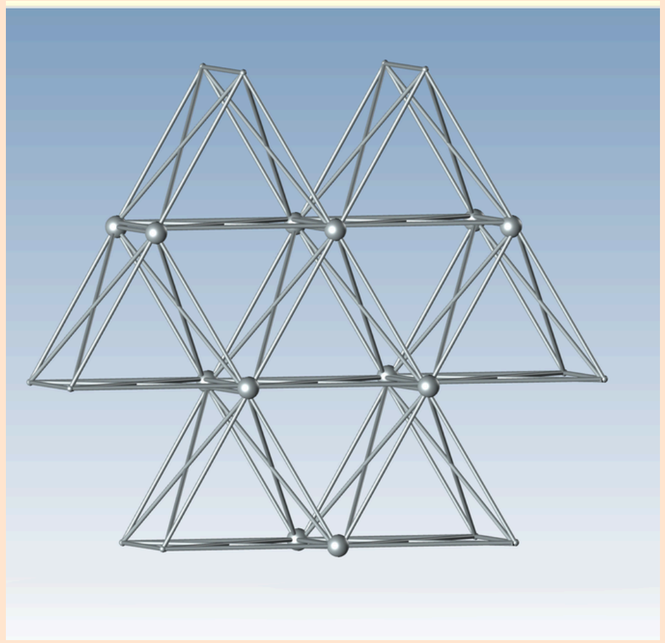


SIDE

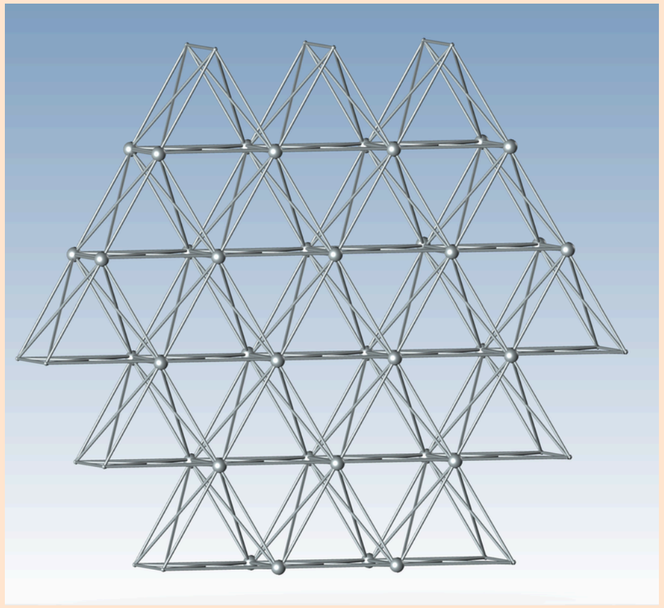
Software **Siemens NX**

Scales **1, 7, and 19 nodes**

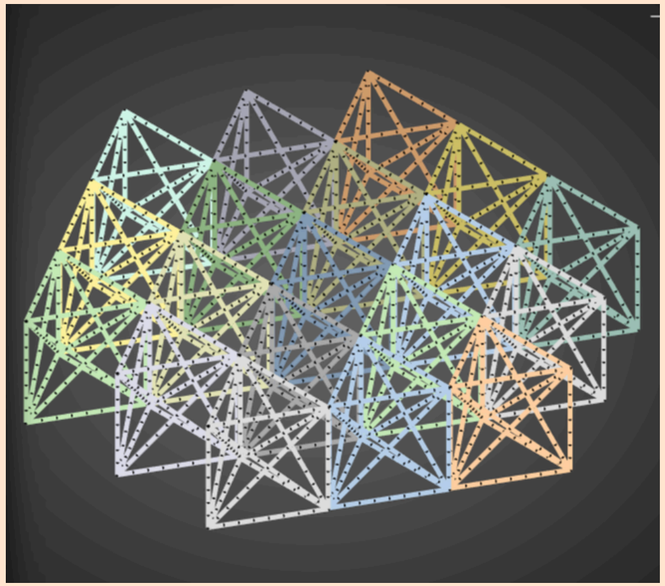
The assembly expanded from 7 to 19 nodes.



7-node model



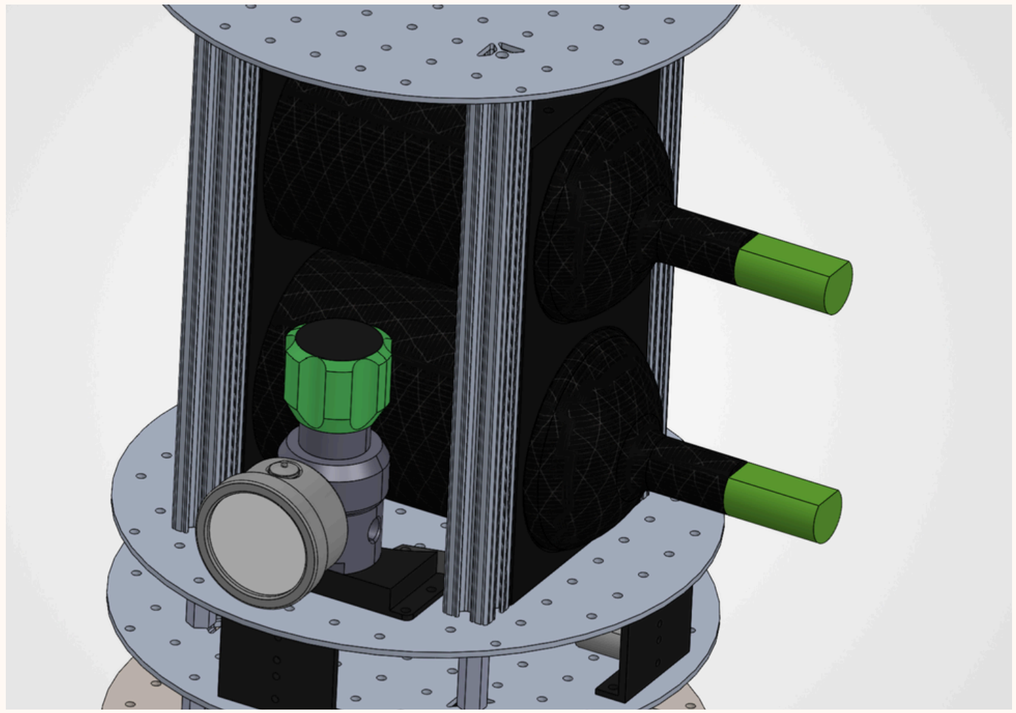
19-node model



Meshed model on Ansys

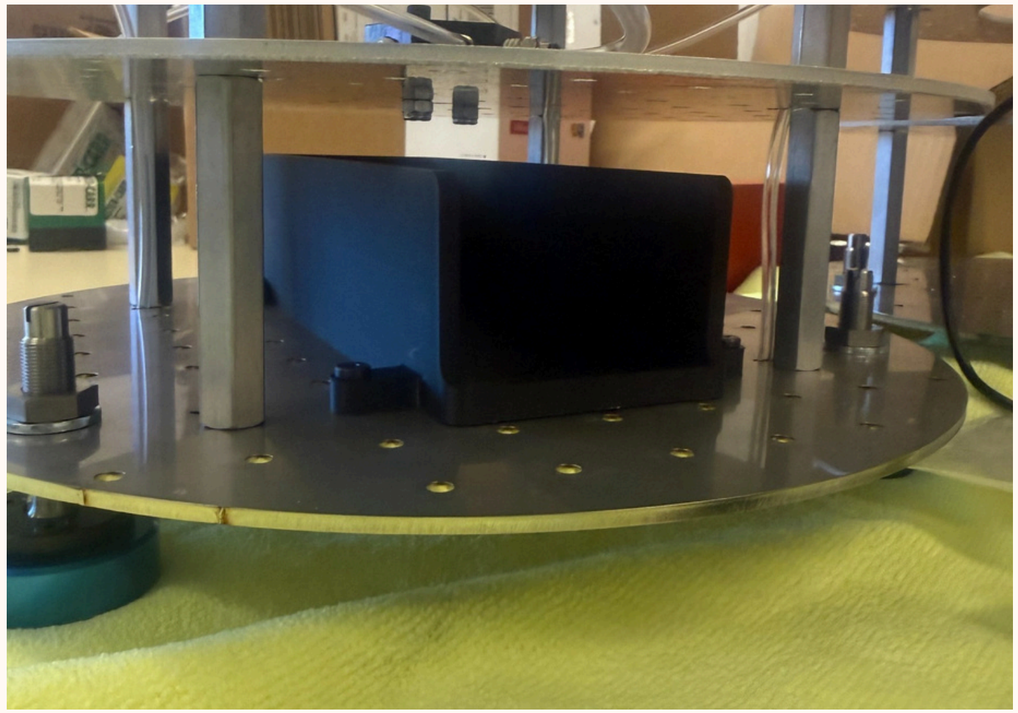
I checked meeting faces and spacing before extending the pattern. The 19-node model was exported to STL, meshed, and linked for FEA frequency analysis.

Impoved on previous Floatbot build



Earlier CAD model

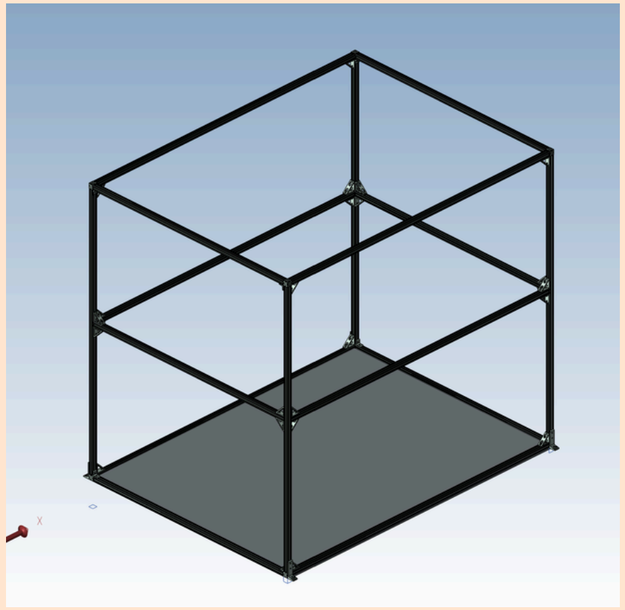
Reworked solenoid valves



New build

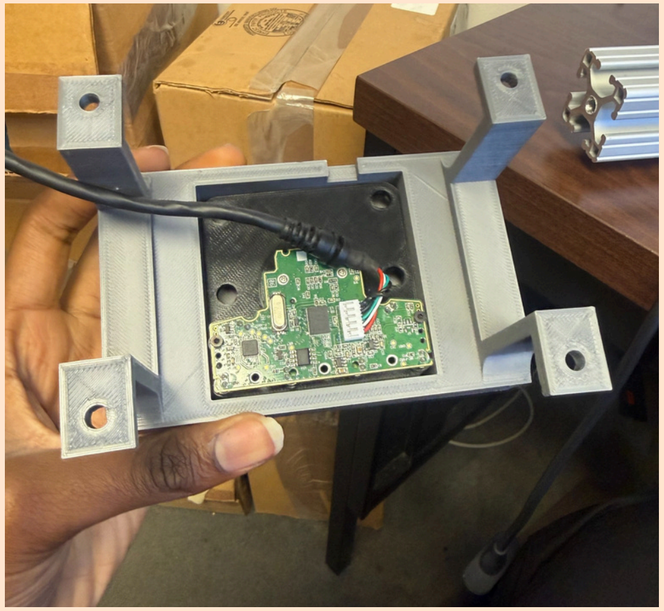
Assembled the next generation floatbots, added 3d printed compoents and did some pdb wiring. Two solenoid valves instead of 1

Floatbot Components



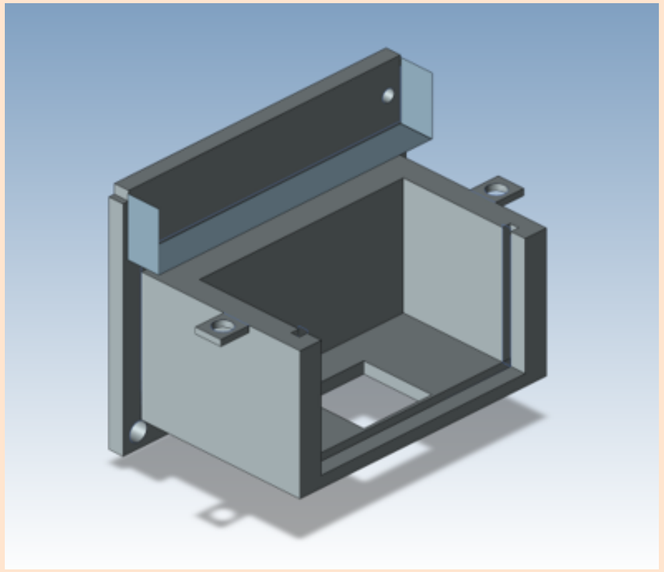
80/20 frame

Stock T-slot brackets and fasteners allowed the frame to be adjusted without welding.



Camera mount- earlier iteration

The base reused existing mounting points. Friction fit to 80/20



Electronics enclosure

One case held the buck converter and PDB with wiring pass-throughs and standoffs.